Software Test Plan (STP)

Project: Online Bookstore

Version: 1.1

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Status: Updated for Mini-Project Deliverable Part-3 (test execution complete except as noted) for Updated for Mini-Project Deliverable Part-3 (test execution complete except as noted) Deliverable 1

A multi-seller e-commerce platform for buying and selling new and second-hand books.

Revision History

Version	Date	Author	Change Summary
0.1	03-Sep-2026	Group 2	Initial test-plan structure adapted from the assigned STP template.
0.5	05-Oct-2026	Group 2	Test scope, strategy, security validation, environment, traceability and test cases aligned with Online Bookstore SRS v1.0.
1.0	05-Oct-2026	Group 2	Baseline STP prepared for Mini-Project Deliverables Part-1.
1.1	06-Nov-2026	Group 2	Part-3 update: execution status and results recorded in Appendix A and B (TC-01 to TC-16); defects DEF-01 and DEF-02 recorded; Appendix C lists the changes made during Sprint 3.

1. Introduction

Purpose: This Software Test Plan defines the testing objectives, scope, strategy, resources, schedule, responsibilities, risks and traceability for the Online Bookstore version 1.0. It is derived from the approved SRS and is intended to guide functional, non-functional, security and acceptance testing.

Scope: Testing covers account and access management; catalogue, search and reviews; cart, wishlist, checkout, payment and orders; seller operations; administrator moderation and reporting; and the performance, security, usability, maintainability and portability requirements stated in the SRS.

Out of scope: real-money settlement, physical logistics integration beyond recording courier tracking information, mobile application development, multi-language content, seller-to-seller bulk trade and machine-learning recommendation features, because these are explicitly outside SRS version 1.0 scope.

References: Online Bookstore SRS v1.0; Online Bookstore SAD v1.0; SRS Requirement Traceability Matrix; Appendix B field layouts and reports; IEEE Std 830-1998; OWASP Top 10 (2021).

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), RTM (Requirement Traceability Matrix), FR (Functional Requirement), NFR (Non-functional Requirement), JWT (JSON Web Token), RBAC (Role-Based Access Control), OTP (One-Time Password).

2. Test Items

- Authentication and user/account module - registration, OTP verification, login/logout, password recovery, profile/address management and RBAC.
- Catalogue and discovery module - catalogue pagination, keyword search, filtering, sorting, book details, reviews and related-title display.
- Cart, wishlist, checkout and order module - cart persistence, quantity validation, checkout revalidation, cancellation and return lifecycle.
- Payment integration - payment intent, signed callback validation, retry handling and idempotent callback processing.
- Seller operations - listing creation/editing, price/MRP validation, inventory, low-stock alerts, dispatch and seller dashboard.
- Administrator operations - moderation, user/listing/review blocking, audit logging and platform analytics/export.
- Notification integration - OTP, order confirmation, invoice and low-stock notifications with retry handling.
- Database and persistence - MongoDB collections, indexes, order-price freezing, ownership and data-integrity rules.
- Redis session/cache behaviour and performance-related caching.
- Responsive React user interface and cross-browser behaviour.

3. Features to be Tested

The following features are mapped directly to the SRS requirement groups. The detailed test cases in Appendix A provide executable coverage.

Requirement group	Features / test focus
FR-01–FR-06	Registration, OTP, authentication, password recovery, profile/address management and RBAC.
FR-07–FR-12	Catalogue paging, indexed search, filtering/sorting, book detail, verified reviews and related titles.
FR-13–FR-18	Cart, wishlist, checkout validation, payment, invoice, order lifecycle, cancellation and returns.
FR-19–FR-24	Seller listings/inventory/fulfilment/dashboard and administrator moderation/analytics.
NFR-01	Search, catalogue/detail and order/report performance under the reference load.
NFR-02	Application availability and health-check monitoring.
NFR-03	Authentication, TLS, payment isolation, input validation, authorization and audit security.
NFR-04	Usability, 360px responsive checkout, WCAG 2.1 AA contrast and keyboard accessibility.
NFR-05	Atomic stock decrement and prevention of overselling.
NFR-06	ESLint quality gate and minimum 70% service-module unit-test coverage.
NFR-07	Compatibility with latest two major versions of Chrome, Edge, Firefox and Safari.
NFR-08	Append-only audit logging for administrative actions and payment events.

4. Features Not to be Tested

- Real-money payment settlement; the SRS restricts payment to the gateway sandbox.
- Storage or processing of card data by the Online Bookstore; card data remains with the payment gateway.
- Physical courier/logistics execution; the system records courier and tracking information only.
- Mobile application implementation; mobile-browser responsiveness is tested, but a native mobile application is out of scope.
- Multi-language content.

- Seller-to-seller bulk trade.
- Machine-learning recommendation engine.
- Dedicated hardware interfaces, because the SRS states that the product uses commodity cloud containers and standard browsers.

5. Test Approach / Strategy

5.1 Test Levels

- Unit Testing: Test individual service modules, validation rules and utility functions in isolation. Service dependencies such as payment gateway and SMTP should be injectable/stubbable.
- Integration Testing: Verify interactions between React/API, Express services/MongoDB, Redis, payment gateway adapter, SMTP adapter and object storage interface.
- System Testing: Execute complete user journeys such as registration → verification → login, search → cart → checkout → payment, seller listing → fulfilment and administrator moderation.
- Acceptance Testing (UAT): Confirm that the implemented system satisfies the demonstrable SRS behaviour and course acceptance expectations.

5.2 Test Types

- Functional testing - verify FR-01 to FR-24.
- Regression testing - rerun impacted tests after defect fixes or changes.
- Performance testing - verify NFR-01 response targets and reference load behaviour.
- Security testing - verify authentication, authorization, TLS, validation, payment isolation, rate limiting and audit controls.
- Usability/accessibility testing - verify NFR-04, responsive behaviour, keyboard access and readable error feedback.
- Compatibility testing - verify NFR-07 across the specified browsers.
- Data-integrity testing - verify NFR-05, including concurrent purchase of the last available copy.

5.3 Test Design Techniques

- Equivalence partitioning for valid/invalid registration, quantity, price, rating and filter inputs.
- Boundary-value analysis for password length, OTP validity, quantity limits, rating 1–5, review length and return-window rules.
- Decision-table testing for RBAC, payment outcomes, order cancellation/return and listing status.
- State-transition testing for order states PLACED → CONFIRMED → PACKED → SHIPPED → DELIVERED and CANCELLED/RETURNED paths.
- Negative testing for unauthorized routes, invalid payment callbacks, invalid ownership and malformed input.

5.4 Entry Criteria

- A testable build is available and deployable.
- Required test data and seeded catalogue are available.
- MongoDB/Redis test environment and external service sandboxes are reachable.
- Critical interfaces have stable contracts sufficient for integration testing.
- Test cases and expected results have been reviewed.

5.5 Exit Criteria

- 100% of planned test cases executed.
- All critical and blocker defects are closed or formally accepted by the project team/faculty.
- All high-priority SRS requirements have at least one passing test.

- Security validation activities in Section 5.1 are completed.
- Performance and data-integrity acceptance targets are demonstrated or documented with results.
- Requirement traceability coverage is complete for the selected test scope.

5.6 Test Case Priority

Priority	Meaning
High	Failure prevents a core buyer/seller/admin journey or violates a critical security/data-integrity requirement.
Medium	Failure affects a supporting function but core transaction flow can continue.
Low	Cosmetic, convenience or lower-impact behaviour.

5.1 Security Validation

Security validation is based on the SRS security requirements and the Online Bookstore security architecture. Testing must demonstrate that security controls are enforced server-side rather than relying only on client-side presentation.

- Authentication validation: verify bcrypt password handling, verified-account requirement, JWT access-token expiry and refresh-token invalidation on logout/password change.
- OTP validation: verify six-digit OTP, ten-minute validity, invalid-attempt handling and request-rate cap of three per hour per account.
- Authorization validation: attempt buyer access to seller/admin endpoints and verify HTTP 403; verify object-level ownership for orders, addresses, carts, listings and settlements.
- TLS validation: verify client-server communication uses TLS 1.2 or higher and plain HTTP is redirected to HTTPS.
- Payment security: verify that card data is never stored/logged by the application; verify payment webhook signature validation and idempotent duplicate-callback handling.
- Input security: test malformed and injection/XSS-oriented inputs against server-side validation and sanitisation.
- Rate limiting: verify repeated failed logins are rate-limited and OTP requests are capped.
- Audit validation: verify administrative actions and payment events record actor, action, target and timestamp in an append-only audit log.
- Session/security review: verify access-token expiry, refresh-token rotation and blocked-user rejection at authentication.

6. Test Environment

Area	Planned environment
Client	React single-page application accessed using the latest two major versions of Chrome, Edge, Firefox and Safari.
Responsive devices	Desktop browser plus mobile-browser viewport down to 360 px as specified by the SRS.
Server	Node.js 20 LTS on a Linux container / managed cloud host.
Database	MongoDB 7 on managed MongoDB Atlas.
Cache / sessions	Redis 7.
Payment	Payment gateway sandbox REST API.
Notification	SMTP service over TLS.
Images	Cloud object storage interface.
API	HTTPS REST API with JSON and bearer JWT.
Test data	Seeded catalogue of 10,000 listings; user roles Buyer/Seller/Administrator; controlled order, payment and inventory records.
Performance load	Reference load of up to 200 concurrent users as stated in SRS.
Tools	GitHub/CI and OpenAPI/Swagger are project-supported tools. Postman and a performance/load tool may be used for execution where available; exact tool selection is an implementation decision.

7. Test Schedule

Milestone	Planned date	Output
Test-plan preparation	05-Oct-2026	STP baseline aligned to SRS/SAD.
Test-case design and review	05-Oct-2026	Functional, non-functional and security test cases.
Environment/data setup	05-Oct-2026	Test environment and seed data ready.
Functional/system execution	06-Oct-2026	Executed test cases and defect records.
Security validation	06-Oct-2026	Security validation results.
Performance/data-integrity validation	06-Oct-2026	NFR-01 and NFR-05 evidence.
Final regression/UAT	07-Oct-2026	Regression results and acceptance evidence.
Test summary	07-Oct-2026	Final Test Summary Report.

8. Test Deliverables

- Software Test Plan (this document).
- Functional, non-functional and security test cases.
- Test data / seeded data required for execution.
- Test execution results and logs.
- Defect reports with severity, evidence and status.
- Requirement Traceability Matrix linking SRS requirements to tests.
- Performance/security validation evidence.
- Final Test Summary Report.

9. Roles and Responsibilities

Role	Name	Responsibility
QA / Test Lead	Deeptha S	Maintain STP, coordinate execution, review coverage and consolidate results.
Test Engineer	Chenchu Sreekrishna Koushik	Design and execute test cases, record evidence and log defects.
Developer / Module Owner	Hardhick M Gowda	Support test environment, analyze defects, implement fixes and provide unit-test evidence.
Product Owner / Faculty Reviewer	Course team / faculty	Review acceptance results and provide final readiness/sign-off.

SRS module ownership used for test coordination: Chenchu Sreekrishna Koushik - Identity/Accounts; Deeptha S - Catalogue/Search/Reviews; G H Pramod - Cart/Checkout/Payment/Orders; Hardhick M Gowda - Seller/Admin/Reporting.

10. Risks and Mitigation

Risk	Impact	Mitigation
Unstable build delivered late	Reduces execution time and coverage.	Request early smoke builds and prioritize high-risk/high-priority test cases.
Payment sandbox unavailable	Checkout success/failure flow cannot be fully executed.	Use controlled test doubles/stubs for service-level tests and execute sandbox tests when available.
SMTP unavailable	OTP and notification tests are blocked.	Stub notification adapter for service tests and retry integration tests when service is restored.
Performance environment differs from reference environment	Measured response times may not be comparable.	Record environment configuration with every performance result and use the SRS reference load.

Insufficient test data	Boundary and ownership tests cannot be executed.	Prepare role-specific users, listings, stock states, orders and payment records before execution.
Defect fixes cause regression	Previously working functionality may break.	Run targeted regression plus the complete high-priority smoke suite after significant fixes.
Third-party free-tier/resource limits	Test environment may become unavailable or slow.	Monitor usage and keep external services within project free-tier constraints.

11. Assumptions & Dependencies

- The Online Bookstore SRS v1.0 remains the baseline for test scope.
- The SAD architecture is implemented sufficiently for integration and system testing.
- MongoDB Atlas and Redis are reachable during test execution.
- The payment gateway sandbox is available and its callback behaviour is sufficiently representative for validation.
- SMTP service is available for integration testing or is replaced by a controlled stub for service-level tests.
- A seeded catalogue of 10,000 listings and a reference load of up to 200 concurrent users are available for performance validation.
- Sellers provide honest book-condition information; the system records the declared condition but cannot physically verify it.
- Courier dispatch itself is outside the system; only courier and tracking information are tested.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than four hours or becomes unusable.
- The build is sufficiently unstable that more than 30% of planned tests are blocked.
- A critical security or data-integrity defect makes further execution unsafe or invalid.
- A required external sandbox dependency is unavailable and no approved test stub is

available. Resume testing if:

- Blocking/critical defects are resolved or an approved workaround is available.
- The test environment and required services are stabilized.
- A new build passes smoke tests.
- Required test data and interfaces are restored.

13. Test Case Management & Traceability

The Requirement Traceability Matrix (RTM) maps SRS requirements to architecture/design references and test cases. The test suite below intentionally covers functional, non-functional and security requirements. The complete project RTM remains the authoritative source for requirement-to-code-to-test mapping.

Requirement	Representative test cases
FR-01–FR-03	TC-01, TC-02, TC-03
FR-06	TC-04
FR-07–FR-10	TC-05, TC-06
FR-11	TC-07
FR-13–FR-16	TC-08, TC-09
FR-18	TC-10
FR-19–FR-21	TC-11
FR-23–FR-24	TC-12
NFR-01	TC-13
NFR-03	TC-03, TC-04, TC-14
NFR-04 / NFR-07	TC-15

NFR-05	TC-16
NFR-08	TC-12, TC-14

14. Test Metrics & Reporting

14.1 Metrics

- % test cases executed = executed test cases / planned test cases × 100.
- % test cases passed and failed.
- Requirement coverage = requirements with at least one executed test / requirements in scope × 100.
- Defect density and defect severity distribution.
- Defect aging - time defects remain open by severity.
- Performance measures including 95th-percentile search response time and order confirmation time.
- Security validation pass/fail status.
- Regression pass rate after defect fixes.

14.2 Reports

- Daily execution status during active testing.
- Defect report / defect dashboard.
- Security validation report.
- Performance test result summary.
- Final Test Summary Report containing execution, pass/fail, defects, coverage and acceptance status.

15. Approvals

Role	Name	Signature / Date
QA / Test Lead	Deeptha S	_____
Development Lead	Hardhick M Gowda	_____
Product Owner / Faculty	_____	_____

Appendix A: System Test Cases

The assignment requires sufficient test cases covering functional and non-functional requirements. The following 16 cases provide broader coverage than the minimum 7–10 requested and are directly aligned with the Online Bookstore SRS.

TC ID	Test Case	SRS Ref.	Priority	Precondition / Action	Expected Result	Status
TC-01	Registration with valid data	FR-01	High	Enter valid buyer registration data.	Account created unverified; six-digit OTP sent.	Executed – Pass
TC-02	OTP verification	FR-02	High	Enter correct OTP within ten minutes.	Account becomes verified and can sign in.	Executed – Pass
TC-03	Login and token handling	FR-03, NFR-03	High	Login with verified valid credentials; inspect expiry/logout.	Access token expires in 15 min; refresh token follows policy; logout invalidates refresh token.	Executed – Pass
TC-04	RBAC unauthorized access	FR-06, NFR-03	High	Call seller/admin endpoint using buyer credentials.	Request rejected with HTTP 403 and security/audit handling occurs.	Executed – Pass
TC-05	Catalogue pagination	FR-07	Medium	Open catalogue with seeded books.	Twenty titles are returned per page with required summary fields.	Executed – Pass
TC-06	Search and filtering	FR-08, FR-09	High	Search by title/author/ISBN/publisher and apply filters.	Correct ranked/filter results returned within performance target.	Executed – Pass
TC-07	Verified-purchase review	FR-11	Medium	Attempt review before delivery, then after DELIVERED.	Pre-delivery review rejected; delivered purchase can submit one 1–5 rating.	Executed – Pass
TC-08	Cart quantity and persistence	FR-13	High	Add listing, change quantity, logout/login, return to cart.	Quantity obeys stock and 1–10 limits; cart persists across sessions.	Executed – Pass
TC-09	Checkout and payment success	FR-15, FR-16, FR-17	High	Checkout valid cart and complete sandbox payment.	Order confirmed only after verified success callback; stock decremented; invoice generated.	Executed – Pass
TC-10	Order cancellation/return	FR-18	High	Cancel before dispatch; attempt return after delivery.	Cancellation accepted before dispatch; return path available within seven days of delivery.	Executed – Pass
TC-11	Seller listing/inventory	FR-19, FR-20, FR-21	High	Create listing, set stock/price, dispatch an order.	Price cannot exceed MRP; low-stock/out-of-stock rules and shipment record work.	Executed – Pass
TC-12	Administrator moderation/reporting	FR-23, FR-24, NFR-08	High	Block listing/user and generate monthly report.	Object is removed from buyer view, reason/action is audited, report generated/exported.	Executed – Pass
TC-13	Search performance	NFR-01	High	Run search under reference load of 10,000 listings/200 concurrent users.	95th-percentile catalogue search is within 2 seconds and indexed query plan is used.	Executed – Pass
TC-14	Security validation suite	NFR-03, NFR-08	High	Test TLS, input validation, rate limiting, payment callback signature and audit events.	Security controls pass; invalid/unauthorized requests are	Executed – Pass

					rejected; events are auditable.	
TC-15	Responsive/accessibility/browser compatibility	NFR-04, NFR-07	Medium	Run buyer flow at 360px and supported browsers; keyboard navigate.	Checkout usable within four screens; controls keyboard reachable; supported browsers function correctly.	Executed – Pass
TC-16	Concurrent last-stock purchase	NFR-05	High	Two concurrent orders attempt to purchase the last copy.	Exactly one succeeds in decrementing the final stock; listing is not oversold.	Executed – Pass

Appendix B: Test Case Execution Record

TC ID	Actual Result	Pass/Fail	Defect ID	Executed By	Date
TC-01	Account created; six-digit OTP sent (mail stub log)	Pass	-	Deeptha	2026-11-03
TC-02	Wrong code rejected; correct code verified; signed in as Buyer2	Pass	-	Deeptha	2026-11-03
TC-03	Access token 15 min; refresh cookie set; refresh 200; logout 204; old refresh token 401	Pass	-	Deeptha	2026-11-03
TC-04	Buyer got 403 FORBIDDEN_ROLE on seller and admin routes; ACCESS_DENIED audit entries written with matching request IDs	Pass	-	Deeptha	2026-11-03
TC-05	20 items per page; page 1 of 125 (2500 books); summary fields present; page 2 returns 20	Pass	-	Deeptha	2026-11-03
TC-06	Title/author/ISBN/ publisher search and filters correct; single search 0.12s (load test under TC-13); observation: search box keeps old text after the search chip is removed (cosmetic)	Pass	-	Deeptha	2026-11-03
TC-07	Review rejected without a delivered purchase of that book record; accepted after delivery (rating 4, count updated); second review for the same order line refused.	Pass	-	Deeptha	2026-11-05
TC-08	Quantity limited by available stock and 1 to 10; cart persisted after logout and login.	Pass	-	Koushik	2026-11-05

TC-09	Order confirmed only after successful payment; stock reduced accordingly.	Pass	-	Koushik	2026-11-05
TC-10	Cancel before dispatch accepted; cancel unavailable after dispatch; return accepted within 7 days of delivery.	Pass	-	Deeptha	2026-11-05
TC-11	Pass after fix. First run found DEF-01 (price above MRP accepted); fixed in PR 87 and retested. Edit, low-stock, out-of-stock and dispatch flow pass.	Pass	DEF-01 (fixed)	Deeptha	2026-11-05
TC-12	Admin blocked a listing with a reason : listing hidden from buyer view, status BLOCKED for seller; block and unblock recorded in audit log with reasons; monthly report exported as CSV and PDF for 2026-11.	Pass	-	Deeptha	2026-11-05
TC-13	200 concurrent connections for 20 s on 10,000 seeded listings, 32k requests, 0 failures: p97.5 234 ms, p99 282 ms (limit 2000 ms). Text index used in the query plan. Per-IP rate limit (SR-10) off via NODE_ENV=test and request logging off for the run; one repeated query so results largely cached; single-machine run.	Pass	-	Deeptha	2026-11-05
TC-14	Login lockout after 5 failures; global limit 100 requests/min (429 after); no/tampered token 401; invalid input, ADMIN self-registration, extra fields and NoSQL operator all 400; webhook without/with wrong signature 401; ACCESS_DENIED, STOCK_EXHAUSTED and payment events present in audit log. HSTS and security headers present; TLS itself not testable on localhost.	Pass	-	Deeptha	2026-11-05
TC-15	Initial run 5 Nov: Add to cart required sideways scrolling at 360 px (DEF-02). Fixed in PR 91. Retest 6 Nov: button visible, Pass.	Pass	DEF-02 (fixed, PR 91)	Koushik	2026-11-06

TC-16	Two buyers placed orders for the last copy and paid concurrently: exactly one order confirmed, the other got INSUFFICIENT_STOCK with refund pending; stock ended at 0 (not negative). First attempt interrupted by the dev auto-reloader; rerun with npm start.	Pass	-	Deeptha	2026-11-05

Appendix: Project Roles and Responsibilities

The following team assignments apply across the Online Bookstore project documents. Requirement ownership in the SRS remains unchanged. The Product Owner / Faculty Reviewer role remains with the course team/faculty.

Role	Assigned member	Responsibilities
QA / Test Lead	Deeptha S	Coordinate test planning and execution, review coverage, consolidate results and track defects.
Developer / Development Lead	Hardhick M Gowda	Coordinate implementation, support the test environment, analyze defects and provide unit-test evidence.
Test Engineer	Chenchu Sreekrishna Koushik	Design and execute test cases, record evidence and log defects.
Requirements & Documentation Coordinator	G H Pramod	Maintain requirements/document consistency, track requirement changes and coordinate project documentation.

This appendix assigns project coordination roles; it does not change SRS requirement owners.

Appendix C: Changes made in Part 3 (Sprint 3)

Test cases were updated to match what was built during Sprint 3. Test case IDs and the RTM rows are unchanged; the requirement-to-test mapping above still holds.

Change in Sprint 3	**Effect on test cases**
Seller page (S3-03), seller listings and orders APIs (S3-01, S3-02)	TC-11 now executed through the seller page and the new APIs (GET /seller/listings, GET /seller/orders).
Mark as delivered (S3-08)	TC-07 and TC-10 use the delivered state set by the seller flow.
Review form on the book page (S3-05)	TC-07 executed through the review form; one review per delivered order line.
Admin moderation and analytics page (S3-04)	TC-12 executed through the admin API; the UI was merged afterwards and is not separately re-executed.
My Orders page (S3-09) and profile / address book page (S3-10)	Covered by the automated API tests; no new system test case added.
Price must not exceed MRP on create (DEF-01, PR 87)	TC-11 retested and passed.
Add to cart and wishlist buttons fit at 360 px (DEF-02, PR 91)	TC-15 retest by the Test Engineer.
Concurrent last-stock script, load test and security checks run against the real application	TC-13, TC-14 and TC-16 executed; limitations recorded in the System Validation Report.